

# Rate floor stable across a $20\times \Delta t$ sweep

exp044 · 2 June 2026

The trained networks this entry uses are produced once in the shared training hub, exp022 (Training), and reused here rather than retrained.

## Abstract

The  $\Delta t$  audit asks whether the exp025 headline E rate is a physical (Hz) property of the trained network or an artefact of the integration timestep. The rate stays within a 9–14 Hz band across a  $20\times \Delta t$  sweep, accuracy holds at 90.4–91.4%, and the gamma cycle period in physical ms is invariant. The rate is not fully  $\Delta t$ -independent, though: it rises monotonically with the timestep, from  $\approx 9.2$  Hz at  $\Delta t = 0.05$  ms to  $\approx 13.4$  Hz at  $\Delta t = 1$  ms, so a coarser step inflates the rate while leaving accuracy and cycle period intact.

## Method

| Parameter                                      | Value                                                                   |
|------------------------------------------------|-------------------------------------------------------------------------|
| Integration timestep $\Delta t$                | 0.05–1.0 ms (swept)                                                     |
| Trial duration $T$                             | 200 ms                                                                  |
| MNIST samples (80/20 stratified split of 7000) | 5600 train / 1400 test ( $\approx 10\%$ of the 70k-sample MNIST corpus) |
| Epochs                                         | 50                                                                      |

The exp022 hub trains one PING per  $\Delta t \in \{0.05, 0.1, 0.25, 0.5, 1.0\}$  ms  $\times$  seed  $\in \{42, 43, 44\}$  = 15 cells; this entry loads them and evaluates. Total physical time  $T = 200$  ms is held constant (step count varies 4000  $\rightarrow$  200). Batch size 64 throughout, smaller than exp025’s 256 but matched across the sweep so per-step compute and memory stay comparable, and the  $\Delta t = 0.05$  cells (4000 timesteps  $\times N_E \times N_I$ ) fit in a single A100. All other PING recipe parameters held to exp025.

Run inference on the test set; report mean E rate (Hz), accuracy, and a single-trial raster from seed 42 per  $\Delta t$  for visual cycle-period inspection.

## Results

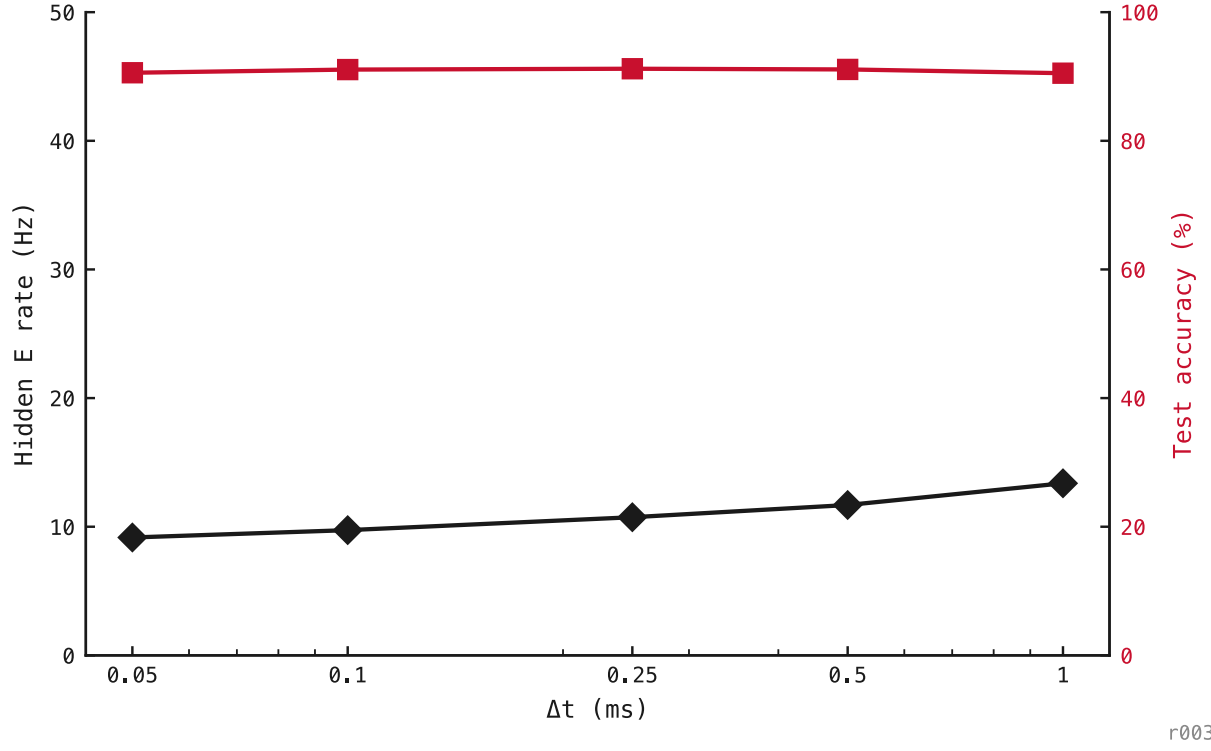


Figure 1: Hidden E rate (black) and test accuracy (red) as  $\Delta t$  varies 20 $\times$ ; markers are per- $\Delta t$  means over three seeds. Accuracy holds near 91% while the E rate rises monotonically from  $\approx 9.2$  Hz at  $\Delta t = 0.05$  ms to  $\approx 13.4$  Hz at  $\Delta t = 1$  ms.

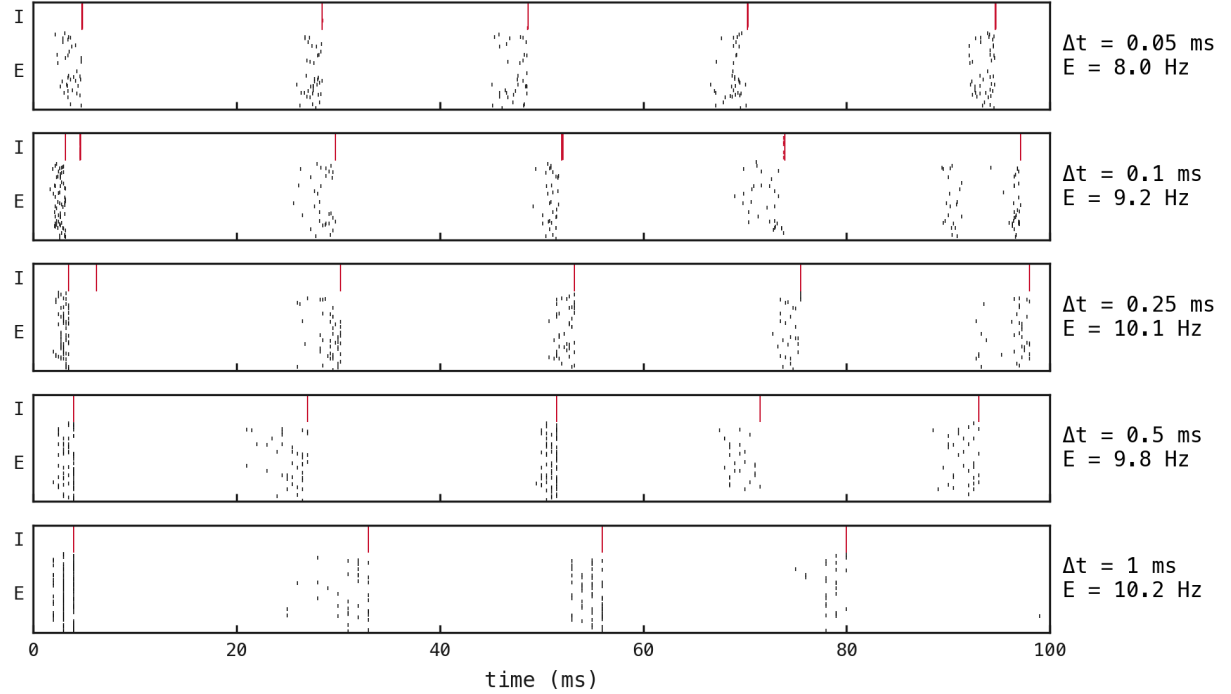
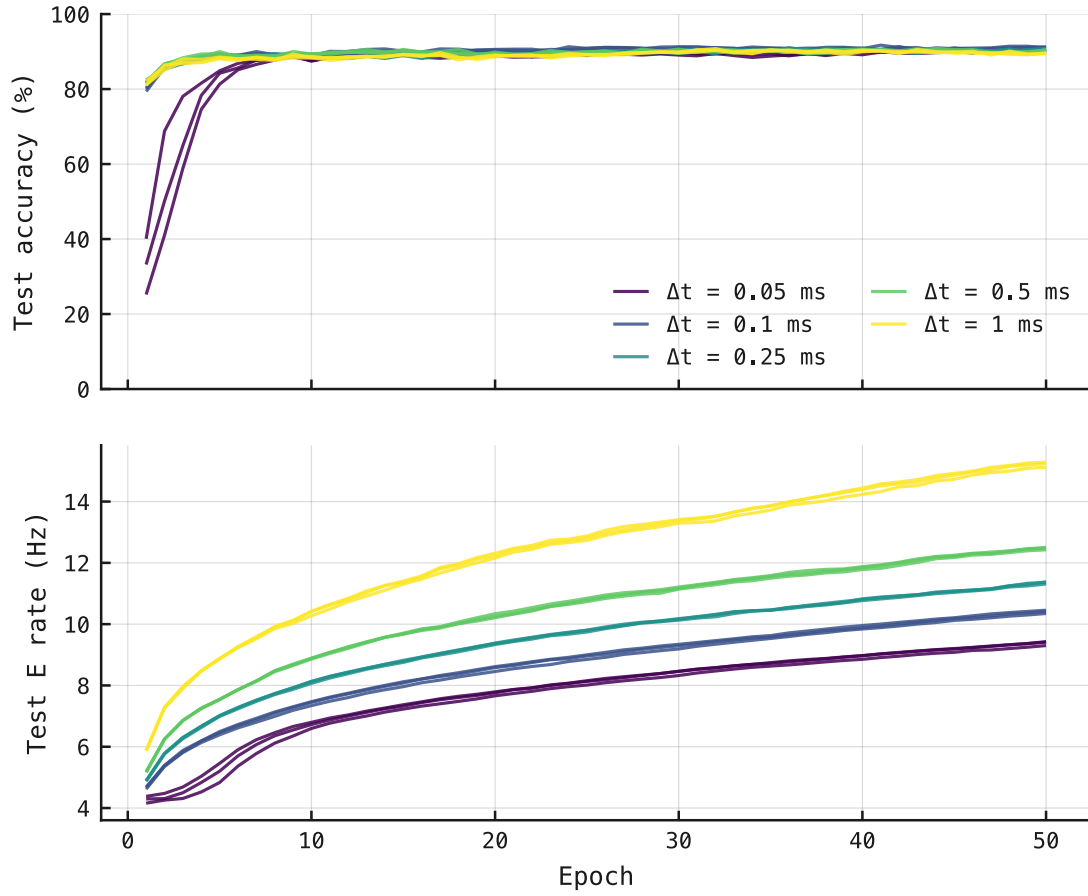


Figure 2: Single-trial rasters at each  $\Delta t$ , x-axis in *physical* ms (not steps). All five panels show E (black) and I (red) bursts locked to the same gamma cadence. The cycle physics is  $\Delta t$ -invariant.



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Figure 3: Top: test accuracy converges by epoch  $\approx 10$ –15 across all  $\Delta t$ . Bottom: test E rate has *not* converged at epoch 50; every curve is still rising, ordered by  $\Delta t$ . The  $\Delta t$ -dependent rate is a snapshot at epoch 50, not a fixed-point ceiling.